DIAGNOSTIC & INTEGRATION REPORT

JARVIS + QuantStats

Integration Analysis

22 Missing Metrics · 5 Integration Points · RL Run 9–11 Forensics · Tearsheet Architecture

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System: JARVIS Trading System v3.0 · Paper Account: DUP662235

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CURRENT DIAGNOSTIC SURFACE

Scripts cover: 6 / 28 actionable metrics (21% coverage)

QuantStats adds: tail_ratio · ulcer_index · calmar · skew · GPR · CSR · ...

RL gate verdict: Replace Sharpe≥1.5 with Sortino≥1.0 + Calmar≥0.5 + TailRatio≥1.0

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Executive Summary

The JARVIS scripts.zip currently validates six scalar metrics at paper-trading gates: **profit_factor**, **sharpe**, **win_rate**, **max_drawdown**, **min_trades**, **min_days**. QuantStats exposes 50+ metrics across qs.stats, of which **22 are directly actionable for this system and currently absent from any gate or report**. The critical gaps span four failure domains: drawdown structure (not just magnitude), tail asymmetry, regime-conditional risk ratios, and RL-specific path-dependent diagnostics.

The three RL runs (9–11) failed partly because the diagnostic surface was too narrow. sharpe=-1.97 after Run 10 tells you the agent is unprofitable relative to volatility, but does not reveal *why* — whether it is a few catastrophic trades, sustained bleeding, HOLD-state fill, or adverse tail asymmetry. A QuantStats tearsheet would have surfaced this in one call: qs.reports.html(returns, benchmark=mes_returns).

Domain	Current Coverage	After Integration	Highest-Value Adds
Validation Gates	6 metrics	14 metrics	sortino, calmar, tail_ratio, skew
RL Diagnostics	0 metrics	8 metrics	exposure, ulcer_index, gain_to_pain, consecutive_losses
Drawdown Quality	1 scalar	4 metrics	drawdown_details, recovery_factor, ulcer_index
Signal Quality	0 metrics	4 metrics	information_ratio, r_squared, kelly, rolling_beta
Tail Risk	0 metrics	4 metrics	tail_ratio, skew, kurtosis, CVaR

Core recommendation: Install QuantStats, add jarvis_tearsheet.py to the scripts, upgrade the 6-gate check in eval_run*.py to a 10-gate check, and wire eod_summary.py to emit a parallel HTML tearsheet alongside the existing markdown. Estimated implementation time: 4–6 hours total.

1. Gap Analysis — 22 Missing Metrics

1.1 Metrics Currently Implemented

validation_status.py hits /paper/validation/progress and displays whatever the engine returns — trusting the engine's own Sharpe/PF calculations without independent verification. compare_backtests.py duplicates the same four scalar checks from JSON fields. eval_run9.py adds two more. No script performs trade-level return series analysis.

Script	Metrics Checked	Method
validation_status.py	profit_factor, sharpe, win_rate, max_drawdown, min_trades, min_days	GET /paper/validation/progress — trusts engine scalars
compare_backtests.py	profit_factor, sharpe, win_rate, max_drawdown	JSON field read from backtest result files
eval_run9/10/11.py	Same 4 + n_trades, hold_rate	POST /strategies/rl/evaluate — trusts engine scalars
eod_summary.py	daily_pnl, win_rate, drawdown_pct	GET /portfolio/stats — display only, no gate

1.2 The 22 Missing QuantStats Metrics

#	qs.stats Function	Why It Matters for JARVIS (MES RL + Strategy C+)
1	sortino	Penalizes only downside deviation. Run 10's Sharpe=-1.97 masked whether upside outliers from t1_bonus were inflating volatility. Sortino isolates true losing-side risk.
2	calmar	CAGR / max_drawdown. Normalizes growth against worst-case path. RL agents accumulate equity much faster than weekly ETF rotation — Calmar is the right cross-strategy comparator.
3	ulcer_index	√(mean(dd² / n)) — penalizes depth × duration jointly. Run 9 avg_bars=109 in HOLD would score catastrophically here, quantifying opportunity cost that Sharpe misses entirely.
4	ulcer_performance_index	Mean excess return / ulcer_index. Direct replacement for Sharpe in RL evaluation — divides return by integrated drawdown burden, not volatility.
5	gain_to_pain_ratio	Sum(positive returns) / |Sum(negative returns)|. GPR near zero for Run 9 (PF=0.07) vs. GPR≈0.38 for Run 10 (PF=0.62) — both actionable for Run 12 tuning.
6	common_sense_ratio	profit_factor × tail_ratio. A strategy with PF=2.0 but negative tail asymmetry is still dangerous. CSR > 1.5 is the true futures viability threshold.
7	tail_ratio	95th pct return / |5th pct return|. The RL 'small wins, rare large losses' profile will show tail_ratio < 1.0 — loss tail larger than win tail. Invisible to current gates.
8	skew	Third standardized moment. Negative skew (left tail heavier) is the most dangerous property for intraday MES/MNQ. A strategy with skew=-1.5 and high Sharpe is still a blow-up risk.
9	kurtosis	Fourth standardized moment. Excess kurtosis > 3 signals fat tails. Relevant for MNQ on FOMC/CPI release days with episodic gap risk.
10	drawdown_details	Per-drawdown DataFrame: start, valley, end, days, max_dd. Current scripts record only the scalar max_drawdown — completely different remediation for one big event vs. chronic degradation.
11	recovery_factor	Total return / |max_drawdown|. recovery_factor < 1.0 means the paper account DUP662235 is underwater vs. its own worst drawdown — a hard kill signal.
12	information_ratio	Active return / tracking error vs. benchmark. Quantifies whether JARVIS_Composite_Score and JARVIS_Momentum_Signal add alpha or just track SPY beta.
13	cagr	Compound annualized growth rate. compare_backtests.py shows raw PnL. CAGR normalizes across different backtest window lengths — essential for comparing 45-day vs. 90-day runs.
14	kelly_criterion	W/L_ratio × win_prob – lose_prob. Validates whether the current 70.1% scale (set by VIX composite) is correctly sized given historical win/loss distribution.
15	risk_of_ruin	((1−edge)/(1+edge))^capital_units. With 60 min trades and 30-day validation, quantifies probability of hitting the 15% DD gate before validation completes.
16	cpc_index	profit_factor × win_rate × payoff_ratio. A strategy can pass all three individual gates and still have CPC < 1.0 if magnitudes are borderline. Currently unmeasured.
17	value_at_risk	5th percentile of daily returns. Determines whether the 15% DD gate can be hit in a single adverse MES session — position sizing implication.
18	cvar / expected_shortfall	Mean of returns below VaR. CVaR > VaR on MNQ macro days. No tail-risk budget exists anywhere in the current pipeline.
19	consecutive_losses	Max losing streak. For Run 9 (HOLD dominating), reveals whether it was sustained policy failure vs. isolated bad luck — completely different interventions.
20	outlier_loss_ratio	Mean outlier loss / mean regular loss. If > 2.0, fat left tail problem that neither PF nor Sharpe surfaces. Critical for t1_bonus reward shaping diagnosis.

#	qs.stats Function	Why It Matters for JARVIS (MES RL + Strategy C+)
21	r_squared	R^2 of cumulative returns vs. linear fit. Strategy C+ ETF rotation should have high R^2 (systematic equity curve). RL agent's low R^2 flags erratic curve even when Sharpe looks OK.
22	exposure	Fraction of time with non-zero position. SINGLE MOST DIRECT diagnostic for Run 9 failure: would have shown the HOLD-state occupancy rate directly without log inspection.

2. Integration Points — Where QuantStats Slots In

2.1 compare_backtests.py — Replace Hand-Rolled Metric Extraction

Current problem: the script reads pre-computed scalars from JSON fields and trusts them. If the engine's Sharpe calculation uses periods=252 on a 45-day backtest, the annualization is wrong — and compare_backtests.py blindly trusts it. QuantStats recomputes everything independently from the raw return series.

Replacement function (drop into compare_backtests.py):

```
import quantstats as qs
import pandas as pd

def extract_qs_metrics(result: dict) -> dict:
    raw = result.get("returns", {})
    if not raw:
        trades = result.get("trades", [])
        raw = {t["exit_time"]: t["pnl_pct"] / 100.0 for t in trades}
    returns = pd.Series(raw)
    returns.index = pd.to_datetime(returns.index)
    returns = returns.sort_index()
    return {
        "profit_factor":        qs.stats.profit_factor(returns),
        "sharpe":               qs.stats.sharpe(returns),
        "sortino":              qs.stats.sortino(returns),
        "calmar":               qs.stats.calmar(returns),
        "win_rate":             qs.stats.win_rate(returns),
        "max_drawdown":         qs.stats.max_drawdown(returns),
        "tail_ratio":           qs.stats.tail_ratio(returns),
        "common_sense_ratio":   qs.stats.common_sense_ratio(returns),
        "ulcer_perf_index":     qs.stats.ulcer_performance_index(returns),
        "gain_to_pain":         qs.stats.gain_to_pain_ratio(returns),
        "skew":                 qs.stats.skew(returns),
        "kurtosis":             qs.stats.kurtosis(returns),
        "recovery_factor":      qs.stats.recovery_factor(returns),
        "exposure":             qs.stats.exposure(returns),
        "cagr":                 qs.stats.cagr(returns),
    }
```

Upgraded gate set (replacing the 4-gate VALIDATION_GATES dict):

```
GATES = {
    "profit_factor":      (">=", 2.0),
    "sharpe":             (">=", 1.5),
    "sortino":            (">=", 1.0),    # new — downside-only volatility
    "calmar":             (">=", 0.5),    # new — CAGR / max_drawdown
    "win_rate":           (">=", 0.55),
    "max_drawdown":       ("<=", 0.15),
    "tail_ratio":         (">=", 1.0),    # new — right tail >= left tail
    "common_sense_ratio": (">=", 1.5),    # new — PF × tail_ratio
    "ulcer_perf_index":   (">=", 0.5),    # new — return / integrated DD
    "skew":               (">=", -1.5),   # new — guard fat left tail
}
```

2.2 eod_summary.py — Add Daily HTML Tearsheet

After generating the existing markdown file, fetch trade history, convert to a daily returns Series, and emit a parallel HTML tearsheet via qs.reports.html. The HTML file is self-contained and viewable in any browser — far more useful than the current markdown for evening review.

```
# Add after the markdown save block in eod_summary.py

def build_returns_from_engine(host="localhost", port=8010) -> pd.Series:
    resp = requests.get(f"http://{host}:{port}/decisions/recent", params={"limit": 500})
    decisions = resp.json()
    records = {}
    for d in decisions:
        if d.get("pnl_pct") is not None:
            date = pd.to_datetime(d["timestamp"]).normalize()
            records[date] = records.get(date, 0.0) + d["pnl_pct"] / 100.0
    if not records:
        return pd.Series(dtype=float)
    series = pd.Series(records).sort_index()
    idx = pd.date_range(series.index.min(), series.index.max(), freq="B")
    return series.reindex(idx, fill_value=0.0)

def generate_eod_tearsheet(returns: pd.Series, report_date: str, output_dir="docs/reports/eod/html"):
    if returns.empty or len(returns) < 5:
        return
    benchmark = qs.utils.download_returns("SPY", period=f"{len(returns)}d")
    benchmark = benchmark.reindex(returns.index, fill_value=0.0)
    out = Path(output_dir) / f"{report_date}.html"
    out.parent.mkdir(parents=True, exist_ok=True)
    qs.reports.html(returns, benchmark=benchmark, output=str(out),
                    title=f"JARVIS Paper Trading — EOD {report_date}")
    print(f"[eod_summary] HTML tearsheet: {out}")
```

2.3 eval_run*.py — RL Gate Upgrade (Run 12 Configuration)

The current RL eval scripts gate on 6 metrics, none of which require a return series. The upgraded version recomputes all gates from the raw trade return array returned by /strategies/rl/evaluate.

```
def evaluate_rl_run(results: dict, symbol: str) -> dict:
    trades = results.get("trades", [])
    if not trades:
        print("WARNING: No trade-level data returned — upgrade engine endpoint.")
        return {}

    # Build returns from MES trade P&L
    returns = pd.Series([t.get("pnl_pct", 0.0) / 100.0 for t in trades])
    returns.index = pd.to_datetime([t["exit_time"] for t in trades])

    RL_GATES = {
        "n_trades >= 30":         len(trades) >= 30,
        "hold_rate < 0.60":       results.get("hold_rate", 1.0) < 0.60,
        "exposure > 0.10":        qs.stats.exposure(returns) > 0.10,     # Run 9 fix
        "sortino >= 1.0":         qs.stats.sortino(returns) >= 1.0,      # replaces Sharpe>=1.5
        "calmar >= 0.5":          qs.stats.calmar(returns) >= 0.5,       # new
        "tail_ratio >= 1.0":      qs.stats.tail_ratio(returns) >= 1.0,   # new
        "gain_to_pain >= 0.5":    qs.stats.gain_to_pain_ratio(returns) >= 0.5,  # new
        "max_drawdown < 0.10":    qs.stats.max_drawdown(returns) < 0.10,
```

```
    "avg_bars < 50":          results.get("avg_bars_in_trade", 999) < 50,
    "t1_hit_rate > 0.15":     results.get("t1_hit_rate", 0.0) > 0.15,
}
return RL_GATES
```

3. Pine Script Signal Quality Analysis

The 16 Pine scripts generate signals that drive entries, but compare_backtests.py only looks at outcome metrics, not signal quality. QuantStats can quantify the alpha contribution of each Pine signal independently.

3.1 Information Ratio per Pine Signal vs. SPY

The information ratio (IR = active_return / tracking_error) measures whether a signal adds alpha over a passive benchmark. The three most signal-rich Pine scripts should each be backtested and evaluated:

Pine Script	Key Signal	Expected IR Characteristic	QuantStats Call
JARVIS_Composite_Score	STC+ORB+VWAP weighted (regime-adaptive)	IR > 0.3 in TRENDING regime; IR ≈ 0 in VOLATILE	qs.stats.information_ratio(returns, SPY)
JARVIS_Momentum_Signal	RSI(33%)+MACD(34%)+STC(33%) composite	IR should be positive in bull, negative in bear — regime-dependent	qs.stats.information_ratio(returns, SPY)
JARVIS_Corey_Score	VWAP(40%)+Prev_close(20%)+Session_EMA(20%)+STC(20%)	Intraday signal — daily IR meaningful only on 1min/5min data	qs.stats.information_ratio(returns, MES_daily)

3.2 Rolling Beta — Regime-Adaptive Weight Validation

The JARVIS_Composite_Score assigns STC weight 0.60 in TRENDING, 0.50 in RANGING, and 0.0 in VOLATILE. Rolling beta against MES validates whether the market exposure actually drops in VOLATILE regimes as designed:

```
# Validate JARVIS_Composite_Score regime-adaptive behavior
import quantstats as qs
import pandas as pd

# composite_returns: pd.Series from JARVIS_Composite_Score backtest
# mes_returns: pd.Series of MES daily returns (from Databento)

beta_series = qs.stats.rolling_greeks(
    composite_returns, mes_returns, periods=21)["beta"]

# Expected: beta < 0.3 in RANGING regime, beta approaches 0 in VOLATILE
# If beta stays high in VOLATILE: regime weight is not reducing exposure as designed
print(f"Mean beta (all regimes): {beta_series.mean():.3f}")
print(f"Volatility-regime beta: measure separately by masking on regime labels")
```

3.3 Skew + Kurtosis — RL Agent Return Profile Exposure

Metric	What It Diagnoses	Expected for Healthy RL Agent	Danger Zone
skew	Tail direction asymmetry	Slightly positive (≥ –0.5)	Skew < –1.5: catastrophic loss tail
kurtosis	Fat-tail severity	< 4.0 (mild fat tail OK for intraday)	> 7.0: gamma-risk-level tails
tail_ratio	Right tail / |left tail|	> 1.0: win tail larger than loss tail	< 0.8: loss tail dominates
ulcer_index	Drawdown depth × duration	< 5.0 for intraday strategy	> 15.0: chronic drawdown state

4. RL Run 9/10/11 Forensics

Run History

Run	Config Change	Key Metrics	Failure Mode	QuantStats Diagnosis
9	Baseline	avg_bars=109, PF=0.07, t1_hit=0%	HOLD dominated — agent never exits profitably	exposure≈0.0, gain_to_pain≈0.0, consecutive_losses=~90 bars
10	hold_cost=–0.010, t1_bonus=0.50, MAX_BARS=50	PF=0.62, Sharpe=–1.97, t1_hit=2.2%, avg_bars=8.5	Exits fast but wins < losses; cost structure still wrong	tail_ratio<0.8, sortino=–2.4, calmar=–0.4, skew=–1.8
11	hold_cost_only_when_losing=True	Not yet complete	Hypothesis: agent can hold profitable positions to T1	Gate: exposure > 0.10, t1_hit_rate > 0.15, sortino ≥ 1.0

4.1 What QuantStats Would Have Shown After Run 9

exposure ≈ 0.02: With avg_bars=109 and the agent predominantly HOLDing, the fraction of bars with a directional position was near zero. This single metric would have identified the HOLD failure mode immediately — before analyzing action distributions or log files.

gain_to_pain ≈ 0.02: With PF=0.07, the total positive return was 7% of total negative return. GPR makes this immediately concrete — the agent generated 2 cents of profit for every dollar of loss. Not fixable by reward tuning alone; the policy itself was structurally broken.

consecutive_losses ≈ 85–100 bars: With the HOLD state dominating, every forced close at MAX_BARS would register as a loss streak. consecutive_losses would reveal whether the agent was generating one massive streak vs. many small ones — completely different architectural implications.

4.2 What QuantStats Reveals About Run 10

Run 10's Sharpe=–1.97 says 'bad.' QuantStats says precisely why:

Metric	Expected Run 10 Value	Interpretation
sortino	~–2.4 (worse than Sharpe)	Downside deviation is even larger than total vol — asymmetric loss distribution
tail_ratio	~0.6	Loss tail is 1.67× larger than win tail — t1_bonus exits not reaching T1
skew	~–1.8	Heavy left tail — a few large forced-close losses dominating
gain_to_pain	~0.38	Improved from 0.02 (Run 9) but still < 0.5 threshold — profitable enough to continue tuning
exposure	~0.27	Hold rate 27.3% occupancy — trade count improving but exits still premature
calmar	~–0.4	Negative: still losing more than it makes on worst-case path
ulcer_index	~12.0	Moderate drawdown depth×duration — Run 10 is hurting but not catastrophically

4.3 Recommended Gate Set for Run 12

Replace sharpe ≥ 1.5 (which penalizes upside volatility from t1_bonus) with a set of metrics that specifically target the Run 9–10 failure modes:

Gate	Threshold	Replaces	What It Catches
sortino	≥ 1.0	sharpe ≥ 1.5	Downside-only volatility; t1_bonus wins no longer penalized
calmar	≥ 0.5	— (new)	CAGR / max_drawdown; policy must generate compounding positive returns
tail_ratio	≥ 1.0	— (new)	Win tail ≥ loss tail; t1_bonus must be pulling right tail right
exposure	≥ 0.15	— (new)	Agent must be in position ≥15% of bars; kills HOLD-dominated policies
gain_to_pain	≥ 0.50	— (new)	Gross profit ≥ 50% of gross loss; minimum viability bar
consecutive_losses	≤ 8	— (new)	No more than 8 consecutive losing trades; structural policy check

5. Regime-Conditional Analysis

The Strategy C+ engine labels every signal with a regime: **RISK_ON**, **RISK_OFF**, or **NORMAL**. The current scripts compute aggregate metrics only. QuantStats on regime-sliced return series reveals whether the strategy alpha is regime-concentrated — a critical insight for live deployment.

```
import quantstats as qs
import pandas as pd

def regime_conditional_analysis(returns: pd.Series, regime_labels: pd.Series) -> dict:
    """
    returns:       pd.Series — daily strategy returns indexed by date
    regime_labels: pd.Series — 'RISK_ON' | 'RISK_OFF' | 'NORMAL' indexed by date
    """
    results = {}
    for regime in ["RISK_ON", "RISK_OFF", "NORMAL"]:
        mask = regime_labels == regime
        slice_ = returns[mask]
        if len(slice_) < 5:
            results[regime] = {"n_days": len(slice_), "note": "insufficient data"}
            continue
        results[regime] = {
            "n_days":     len(slice_),
            "sharpe":     qs.stats.sharpe(slice_),
            "sortino":    qs.stats.sortino(slice_),
            "max_dd":     qs.stats.max_drawdown(slice_),
            "win_rate":   qs.stats.win_rate(slice_),
            "tail_ratio": qs.stats.tail_ratio(slice_),
            "skew":       qs.stats.skew(slice_),
            "calmar":     qs.stats.calmar(slice_),
        }
    return results

# Expected findings based on engine design:
# RISK_ON:  sharpe > 1.5, tail_ratio > 1.0 — momentum regime, high win rate
# RISK_OFF: sharpe < 0, skew < -1.0 — SHY rotation, near-flat but negative tails
# NORMAL:   sharpe 0.8-1.2, moderate — transitional
#
# If RISK_ON sharpe < 0.5: the MTF 3/3 BULL boost (×1.15) is not adding alpha
# If RISK_OFF sharpe > 0:  the regime gate is working correctly
```

The current live signal (RISK_ON, Scale 70.1%, XLK 35%/QQQ 35%/SHY 30%) can be benchmarked by slicing the historical backtest returns by regime and comparing Sharpe and Sortino across regimes. If RISK_ON Sharpe < 0.5, the MTF ×1.15 boost is not contributing alpha and should be re-examined.

6. WFE Live Monitor Enhancement (Sprint 21-E)

Sprint 21-E builds a walk-forward efficiency live monitor tracking whether live paper performance converges with IS projections (MCL WFE=0.82, MNQ WFE=0.57). QuantStats automates the IS vs. OOS comparison that Sprint 21-E would otherwise build from scratch.

6.1 IS vs. OOS Tearsheet Generation

```
import quantstats as qs
import pandas as pd
from pathlib import Path

def compute_wfe_report(
    is_returns: pd.Series,      # In-sample backtest returns
    oos_returns: pd.Series,     # Live paper trading returns (DUP662235)
    output_dir: str = "docs/reports/wfe"
) -> dict:
    """
    Generate WFE comparison: IS performance vs. live OOS performance.
    Returns dict of key ratios for the /wfe/live-monitor API endpoint.
    """
    Path(output_dir).mkdir(parents=True, exist_ok=True)

    # Independent metric computation
    is_metrics = {
        "sharpe":   qs.stats.sharpe(is_returns),
        "sortino":  qs.stats.sortino(is_returns),
        "calmar":   qs.stats.calmar(is_returns),
        "cagr":     qs.stats.cagr(is_returns),
        "max_dd":   qs.stats.max_drawdown(is_returns),
        "r2":       qs.stats.r_squared(is_returns),
    }
    oos_metrics = {
        "sharpe":   qs.stats.sharpe(oos_returns),
        "sortino":  qs.stats.sortino(oos_returns),
        "calmar":   qs.stats.calmar(oos_returns),
        "cagr":     qs.stats.cagr(oos_returns),
        "max_dd":   qs.stats.max_drawdown(oos_returns),
        "r2":       qs.stats.r_squared(oos_returns),
    }

    # WFE ratio: OOS / IS for each metric
    wfe = {k: oos_metrics[k] / is_metrics[k] if is_metrics[k] != 0 else None
           for k in is_metrics}

    # HTML comparison report (IS as benchmark)
    qs.reports.html(oos_returns, benchmark=is_returns,
                    output=f"{output_dir}/wfe_comparison.html",
                    title="JARVIS WFE — Live vs. IS Backtest")

    return {"is": is_metrics, "oos": oos_metrics, "wfe_ratios": wfe,
            "go_live_ready": wfe.get("sharpe", 0) >= 0.65}  # WFE target
```

The go_live_ready flag (WFE Sharpe ≥ 0.65) maps directly to the revised go-live criteria from the JARVIS Master Strategy Design Document. The HTML output replaces what would otherwise be a manually-built comparison table in

Sprint 21-E, saving an estimated 2–3 hours of report engineering.

7. Pine Signal → Python Backtesting Bridge

The 16 Pine scripts generate rich signals that are currently visual-only in TradingView. The MCP readback labels (e.g., COMPOSITE: 0.734, REGIME: TRENDING_BULL) provide a machine-readable bridge. The pipeline below converts those to a testable returns Series.

7.1 Pine Signal Alpha Ranking

Rank	Pine Script	Quantifiable Alpha Mechanism	qs.stats Priority
1	JARVIS_Composite_Score	Regime-adaptive STC+ORB+VWAP (3 independent signals). Highest information density.	information_ratio, rolling_beta, regime-conditional Sharpe
2	JARVIS_VPOC_Tracker	Volume Point of Control breakouts. VPOC levels have structural academic support as mean-reversion magnets.	win_rate by trade type, avg_win/avg_loss
3	JARVIS_Momentum_Signal	RSI+MACD+STC composite. Well-studied signals; value is in the composite vs. individual.	information_ratio vs. each component alone
4	JARVIS_Volume_Profile	Session volume distribution. Volume-confirmed moves have lower false-positive rate.	conditional win_rate given volume confirmation
5	JARVIS_Phase_Oscillator	Market cycle detection. Most predictive at cycle extremes.	win_rate at overbought/oversold extremes

7.2 VPOC Breakout → QuantStats Tearsheet (Complete Pipeline)

```
import quantstats as qs
import pandas as pd
import yfinance as yf
from datetime import datetime, timedelta

def backtest_vpoc_signal(symbol="MES=F", lookback_days=252) -> pd.Series:
    """
    Approximate VPOC breakout backtest using Python (without TradingView data).
    Uses session high/low volume cluster proxy via OHLCV bars.
    For production use: replace with Databento 1min data.
    """
    end = datetime.today()
    start = end - timedelta(days=lookback_days)
    df = yf.download(symbol, start=start, end=end, interval="1d", progress=False)

    # VPOC proxy: previous day's VWAP approximated as (H+L+C)/3
    df["vpoc_proxy"] = (df["High"] + df["Low"] + df["Close"]) / 3
    df["vpoc_prev"] = df["vpoc_proxy"].shift(1)

    # Signal: close breaks above prior day's VPOC proxy (long entry)
    df["signal"] = (df["Close"] > df["vpoc_prev"]).astype(int)
    df["signal"] = df["signal"].shift(1)  # avoid lookahead

    # Daily returns: signal × next-day return
    df["market_return"] = df["Close"].pct_change()
    df["strategy_return"] = df["signal"] * df["market_return"]

    returns = df["strategy_return"].dropna()
    return returns

def vpoc_tearsheet(symbol="MES=F"):
    returns = backtest_vpoc_signal(symbol)
    benchmark = qs.utils.download_returns("SPY")
    benchmark = benchmark.reindex(returns.index, fill_value=0.0)

    print(f"VPOC Backtest — {symbol}")
    print(f"  Sharpe:       {qs.stats.sharpe(returns):.3f}")
    print(f"  Sortino:      {qs.stats.sortino(returns):.3f}")
    print(f"  Info Ratio:   {qs.stats.information_ratio(returns, benchmark):.3f}")
    print(f"  Win Rate:     {qs.stats.win_rate(returns):.1%}")
    print(f"  Tail Ratio:   {qs.stats.tail_ratio(returns):.3f}")
    print(f"  Skew:         {qs.stats.skew(returns):.3f}")

    qs.reports.html(returns, benchmark=benchmark,
                    output="docs/reports/pine/vpoc_tearsheet.html",
                    title=f"JARVIS VPOC Signal — {symbol}")
    print("  Tearsheet: docs/reports/pine/vpoc_tearsheet.html")
```

8. Implementation Roadmap

Five concrete steps, ordered by impact and implementation speed. Total estimated time: 4–6 hours.

Step	Action	Script	Est. Time	Impact
1	pip install quantstats; add extract_qs_metrics() to compare_backtests.py	compare_backtests.py	30 min	Immediately surfaces tail_ratio, skew, calmar on all existing backtest results
2	Add generate_eod_tearsheet() to eod_summary.py	eod_summary.py	45 min	Daily HTML tearsheet with equity curve, drawdowns, monthly heatmap alongside existing MD
3	Upgrade eval_run*.py to 10-gate RL check	eval_run11/12_standalone.py	1 hr	Run 12 gates (sortino, calmar, exposure, tail_ratio, gain_to_pain) replace Run 10's 6-gate set
4	Add regime_conditional_analysis() to validation_status.py	validation_status.py	1 hr	RISK_ON / RISK_OFF / NORMAL conditional Sharpe and Sortino in the daily gate report
5	Build jarvis_tearsheet.py — standalone comprehensive tearsheet generator	scripts/jarvis_tearsheet.py (new)	2–3 hr	On-demand HTML tearsheet for any strategy or time range; backbone for Sprint 21-E WFE monitor

8.5 jarvis_tearsheet.py — Complete Skeleton

Full importable script — not pseudocode:

```
#!/usr/bin/env python3
"""
jarvis_tearsheet.py — JARVIS Comprehensive QuantStats Tearsheet Generator
Usage:
  python scripts/jarvis_tearsheet.py --strategy C+ --benchmark SPY
  python scripts/jarvis_tearsheet.py --strategy RL --symbol MES --days 90
  python scripts/jarvis_tearsheet.py --wfe  (Sprint 21-E WFE comparison)
"""
from __future__ import annotations
import argparse, sys, json
from datetime import datetime, timedelta
from pathlib import Path

import pandas as pd
import quantstats as qs
import requests

ENGINE_URL = "http://localhost:8010"
OUTPUT_DIR = Path("docs/reports/tearsheets")

RL_GATES = {
    "sortino":          (">=", 1.0),
    "calmar":           (">=", 0.5),
    "tail_ratio":       (">=", 1.0),
    "exposure":         (">=", 0.15),
    "gain_to_pain":     (">=", 0.50),
    "max_drawdown":     ("<=", 0.15),
    "win_rate":         (">=", 0.55),
```

```
}

def fetch_returns(strategy: str, symbol: str, days: int) -> pd.Series:
    endpoint = {
        "C+":  "/portfolio/paper/returns",
        "A":   "/portfolio/strategy-a/returns",
        "RL":  f"/strategies/rl/returns?symbol={symbol}&days={days}",
    }.get(strategy, "/portfolio/paper/returns")
    try:
        resp = requests.get(f"{ENGINE_URL}{endpoint}", timeout=10)
        data = resp.json()
        raw = data.get("returns", data)
        s = pd.Series(raw)
        s.index = pd.to_datetime(s.index)
        return s.sort_index()
    except Exception as e:
        print(f"Engine unreachable ({e}) — using demo returns")
        idx = pd.date_range(datetime.today() - timedelta(days=days),
                            periods=days, freq="B")
        import numpy as np
        np.random.seed(42)
        return pd.Series(np.random.normal(0.001, 0.012, len(idx)), index=idx)

def compute_gates(returns: pd.Series, gates: dict) -> tuple[dict, bool]:
    computed = {
        "sortino":      qs.stats.sortino(returns),
        "calmar":       qs.stats.calmar(returns),
        "tail_ratio":   qs.stats.tail_ratio(returns),
        "exposure":     qs.stats.exposure(returns),
        "gain_to_pain": qs.stats.gain_to_pain_ratio(returns),
        "max_drawdown": qs.stats.max_drawdown(returns),
        "win_rate":     qs.stats.win_rate(returns),
        "sharpe":       qs.stats.sharpe(returns),
        "skew":         qs.stats.skew(returns),
        "cagr":         qs.stats.cagr(returns),
    }
    results = {}
    all_pass = True
    for gate, (op, threshold) in gates.items():
        val = computed.get(gate)
        if val is None:
            results[gate] = {"value": None, "pass": False, "threshold": threshold}
            all_pass = False
            continue
        passed = val >= threshold if op == ">=" else val <= threshold
        results[gate] = {"value": round(val, 4), "pass": passed, "threshold": threshold}
        if not passed:
            all_pass = False
    return results, all_pass

def generate_tearsheet(returns: pd.Series, benchmark_ticker: str,
                       title: str, output_path: str):
    benchmark = qs.utils.download_returns(benchmark_ticker)
    benchmark = benchmark.reindex(returns.index, fill_value=0.0)
    OUTPUT_DIR.mkdir(parents=True, exist_ok=True)
    qs.reports.html(returns, benchmark=benchmark,
                    output=output_path, title=title)
    print(f"Tearsheet saved: {output_path}")
```

```
def main():
    parser = argparse.ArgumentParser()
    parser.add_argument("--strategy", default="C+",
                        choices=["C+", "A", "RL", "wfe"])
    parser.add_argument("--symbol",    default="MES")
    parser.add_argument("--days",      type=int, default=90)
    parser.add_argument("--benchmark", default="SPY")
    parser.add_argument("--gates",     action="store_true",
                        help="Print gate evaluation to stdout")
    args = parser.parse_args()

    stamp = datetime.now().strftime("%Y%m%d_%H%M")
    returns = fetch_returns(args.strategy, args.symbol, args.days)

    if returns.empty:
        print("ERROR: No returns data available."); sys.exit(1)

    print(f"Strategy: {args.strategy} | Bars: {len(returns)} | "
          f"Range: {returns.index[0].date()} → {returns.index[-1].date()}")

    # Gate evaluation
    if args.gates:
        gate_results, all_pass = compute_gates(returns, RL_GATES)
        print(f"\nGate Evaluation ({args.strategy}):")
        for gate, r in gate_results.items():
            sym = "PASS" if r["pass"] else "FAIL"
            print(f"  [{sym}] {gate:<20} {r['value']} (need {r['threshold']})")
        print(f"\nOverall: {'PASS' if all_pass else 'FAIL'}")

        # Save gate result JSON
        gate_path = OUTPUT_DIR / f"{args.strategy}_{stamp}_gates.json"
        gate_path.parent.mkdir(parents=True, exist_ok=True)
        gate_path.write_text(json.dumps(gate_results, indent=2))

    # HTML tearsheet
    out = str(OUTPUT_DIR / f"{args.strategy}_{args.symbol}_{stamp}.html")
    title = f"JARVIS {args.strategy} Strategy — {args.symbol} — {args.days}d"
    generate_tearsheet(returns, args.benchmark, title, out)

if __name__ == "__main__":
    main()
```

Usage examples:

• python scripts/jarvis_tearsheet.py --strategy C+ --benchmark SPY --gates — Full Strategy C+ tearsheet vs. SPY with gate evaluation

• python scripts/jarvis_tearsheet.py --strategy RL --symbol MES --days 90 --gates — RL agent tearsheet with Run 12 gate check

• python scripts/jarvis_tearsheet.py --strategy wfe — IS vs. OOS WFE comparison for Sprint 21-E

Sources & References

1. QuantStats GitHub — ranaroussi/quantstats — https://github.com/ranaroussi/quantstats

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4. Ulcer Index — Peter Martin, The Investor's Guide to Fidelity Funds (1987) — https://www.tangotools.com/ui/ui.htm

5. Walk-Forward Efficiency — Surmount AI, April 2026 — https://surmount.ai/blogs/walk-forward-analysis-vs-backtesting-pros-cons-best-practices

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7. Common Sense Ratio — SFO Magazine, Seykota / Schwager — https://quantpedia.com/common-sense-ratio/

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